Status	Finished
Started	Friday, 16 October 2026, 11:52 AM
Completed	Friday, 16 October 2026, 11:55 AM
Duration	3 mins 19 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Linear Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Sorting
c. Searching
d. Rearranging

The correct answer is: Sorting

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Linear Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. Searching
c. All of the above
d. Selection

The correct answer is: All of the above

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Finding a specific element in a list
b. Merging two datasets
c. Removing duplicates from a list
d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **6**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation
b. It increases the number of comparisons needed
c. It makes the search operation slower
d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It sorts data out of the box efficiently
c. It only works with integer arrays
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **8**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It does not require recursion
c. It is based on the divide-and-conquer approach
d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Linear Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Provides a basis for other algorithms to work efficiently
b. Decreases the efficiency of algorithms
c. Makes data harder to manage
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **11**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Searching for data in a list
c. Arranging data in a particular format
d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are discarded
- c. They are compared element by element
- d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Quick sort
- b. Selection sort
- c. Heap sort
- d. Merge sort

The correct answer is:
Merge sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It simplifies the sorting process by using only one pass
- b. It allows for efficient parallel processing and sorting of data
- c. It only works on small datasets
- d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **15**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It makes the analysis slower
- c. It complicates the analysis process
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates